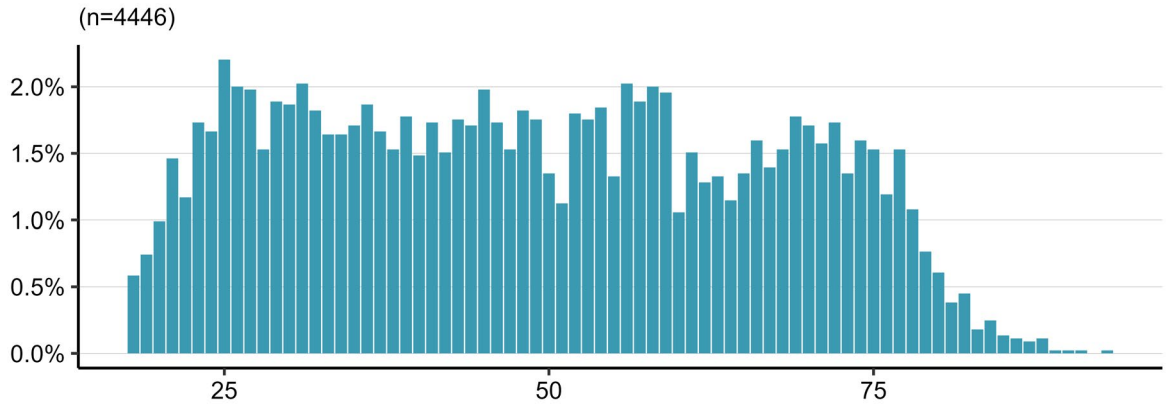
Survey – Real world

The survey, conducted among 4,446 households as part of the iFlex project during the European energy crisis, is detailed in the following paper: *Hofmann, Matthias, Sigurd Bjarghov, and Stian Nessa. 'Norwegian Hourly Residential Electricity Demand Data with Consumer Characteristics during the European Energy Crisis'. Data in Brief 51, no. 109687 (1 December 2023). https://doi.org/10.1016/j.dib.2023.109687.*

The survey data are published at Zenodo: https://zenodo.org/records/8423312

This document is published at Zenodo: https://zenodo.org/records/11580541

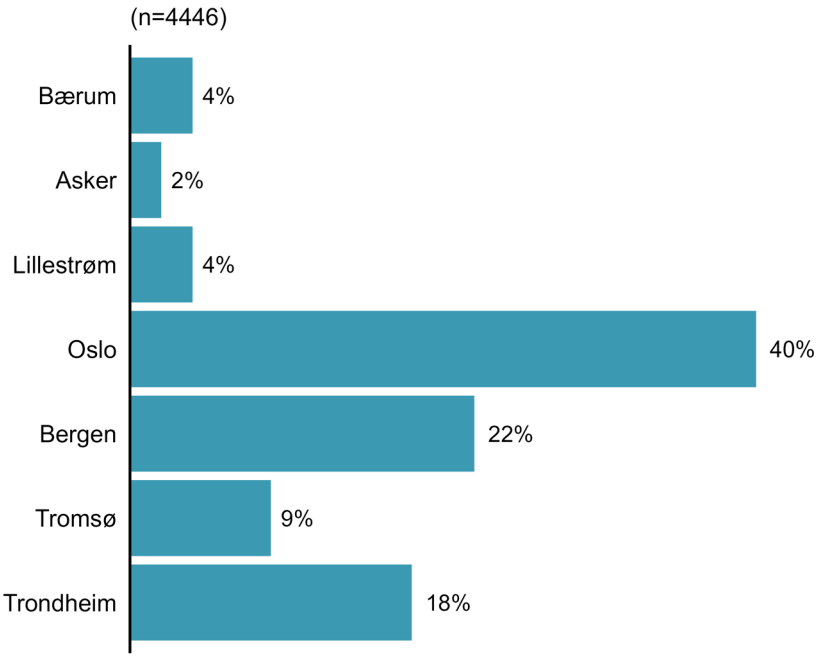
Question Q_Age: Age



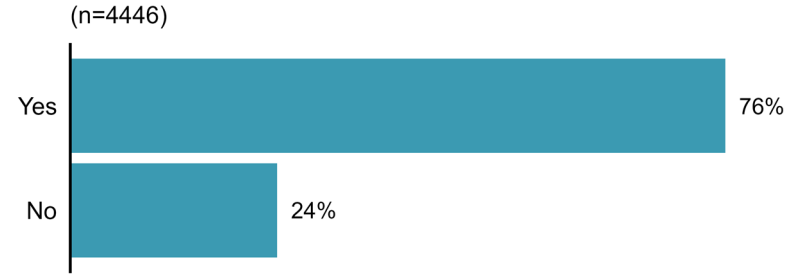
Question Q_Gender: Gender



Question Q_City: City



Question Q1: Did you monitor your power consumption this winter?



(IF Question Q1 = Yes)
Question Q2_1: How did you acquire information about you power consumption? Multiple answers possible: Power bill



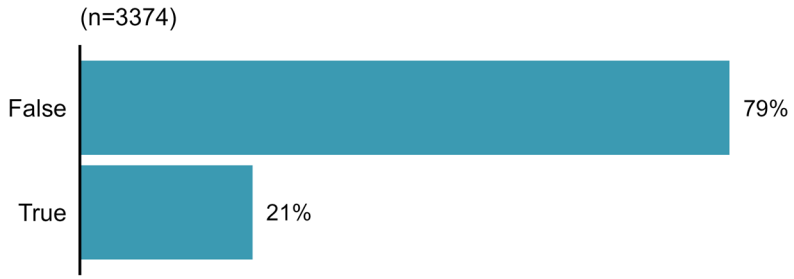
(IF Question Q1 = Yes)
Question Q2_2: How did you acquire information about you power consumption? Multiple answers possible: Display on the smartmeter



(IF Question Q1 = Yes)
Question Q2_3: How did you acquire information about you power consumption? Multiple answers possible: Real time consumption via the HAN-port



(IF Question Q1 = Yes)
Question Q2_4: How did you acquire information about you power consumption? Multiple answers possible: Website (power company, grid company, Elhub, etc.)



(IF Question Q1 = Yes)
Question Q2_5: How did you acquire information about you power consumption? Multiple answers possible: App (power company, grid company, etc.)



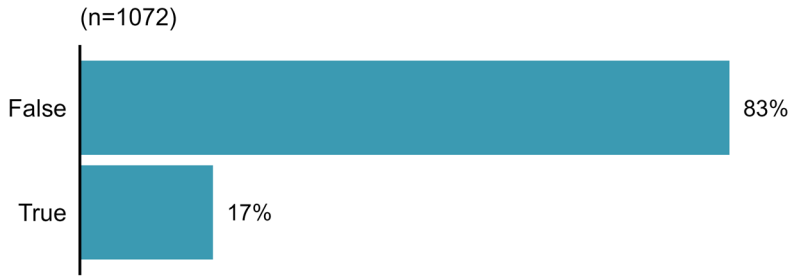
(IF Question Q1 = Yes)
Question Q2_6: How did you acquire information about you power consumption? Multiple answers possible: Email (power company, grid company)



(IF Question Q1 = Yes)
Question Q2_7: How did you acquire information about you power consumption? Multiple answers possible: Other:



(IF Question Q1 = No)
Question Q3_1: Why did you not monitor your consumption? Multiple answers possible: No time or interest in checking consumption



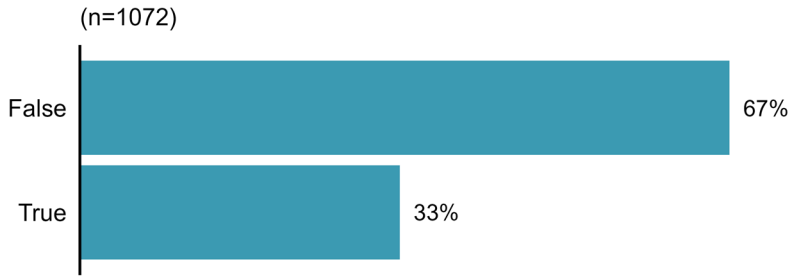
(IF Question Q1 = No)
Question Q3_2: Why did you not monitor your consumption? Multiple answers possible: Difficult finding information about consumption



(IF Question Q1 = No)
Question Q3_3: Why did you not monitor your consumption? Multiple answers possible: Difficult to understand what could have been done



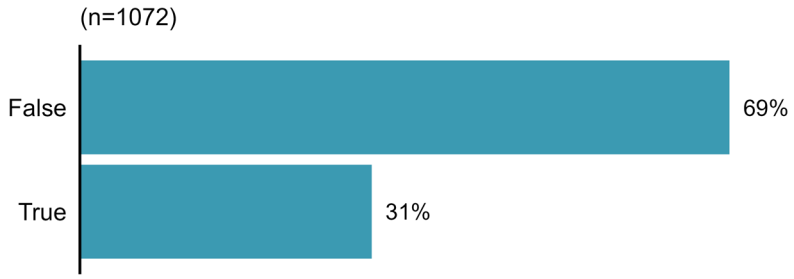
(IF Question Q1 = No)
Question Q3_4: Why did you not monitor your consumption? Multiple answers possible: Few possibilitees to change consumption



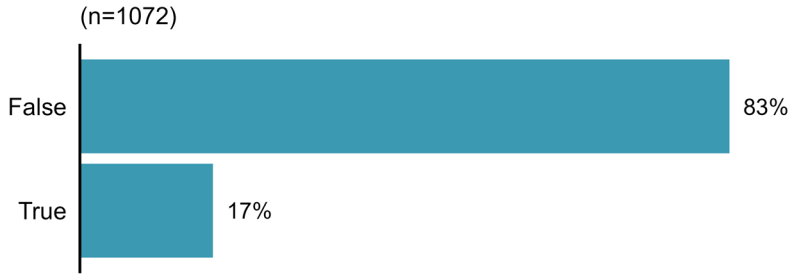
(IF Question Q1 = No)
Question Q3_5: Why did you not monitor your consumption? Multiple answers possible: Potential savings too low



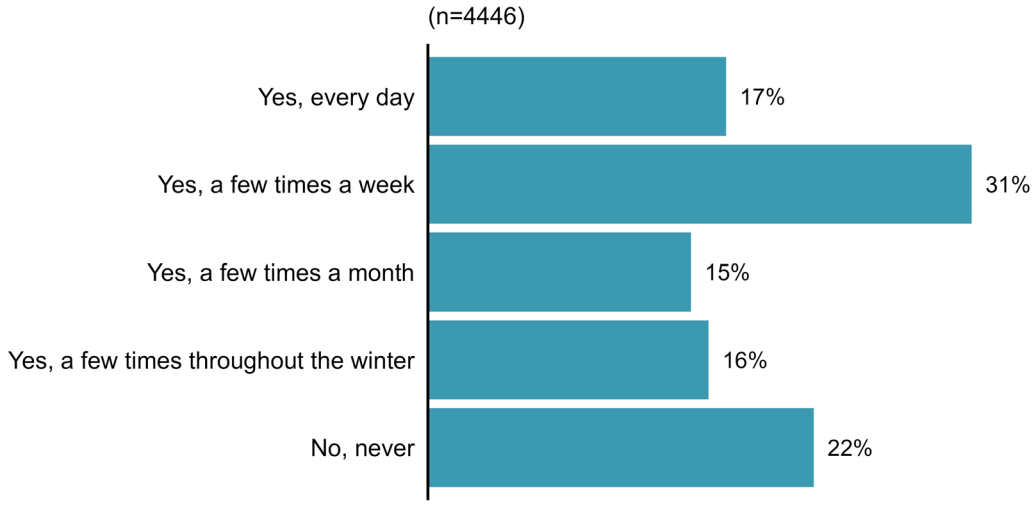
(IF Question Q1 = No)
Question Q3_6: Why did you not monitor your consumption? Multiple answers possible: Power bill too low



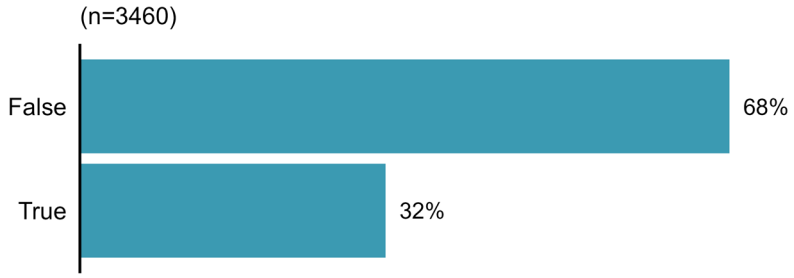
(IF Question Q1 = No)
Question Q3_7: Why did you not monitor your consumption? Multiple answers possible: Other:



Question Q4: Did you monitor the variation in electricity prices from day to day and hour to hour this winter?



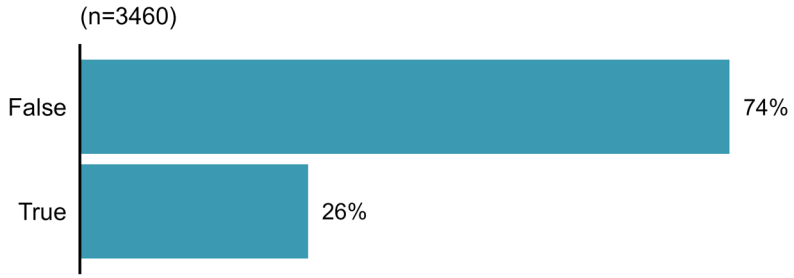
(IF Question Q4 = Yes, …)
Question Q5_1: How did you acquire information about the electricity prices? Multiple answers possible: Power bill



(IF Question Q4 = Yes, …)
Question Q5_2: How did you acquire information about the electricity prices? Multiple answers possible: Media/news



(IF Question Q4 = Yes, …)
Question Q5_3: How did you acquire information about the electricity prices? Multiple answers possible: Website (power company, grid company, NordPool, etc.)



(IF Question Q4 = Yes, …)
Question Q5_4: How did you acquire information about the electricity prices? Multiple answers possible: App (power company, grid company, NordPool, etc.)



(IF Question Q4 = Yes, …)
Question Q5_5: How did you acquire information about the electricity prices? Multiple answers possible: Email (power company, grid company)



(IF Question Q4 = Yes, …)
Question Q5_6: How did you acquire information about the electricity prices? Multiple answers possible: SMS (power company, grid company)



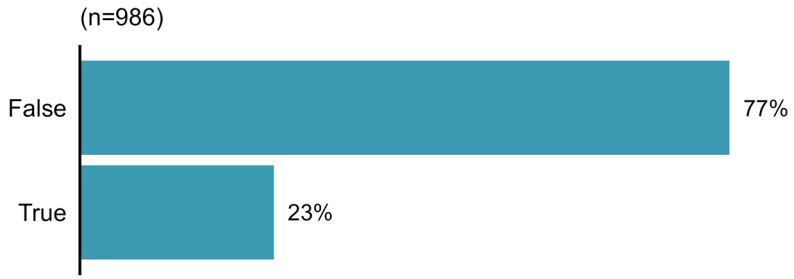
(IF Question Q4 = Yes, …)
Question Q5_7: How did you acquire information about the electricity prices? Multiple answers possible: From work



(IF Question Q4 = Yes, …)
Question Q5_8: How did you acquire information about the electricity prices? Multiple answers possible: Other:



(IF Question Q4 = Yes, …)
Question Q6_1: Why did you not monitor the electricity prices? Multiple answers possible: No time or interest in checking prices



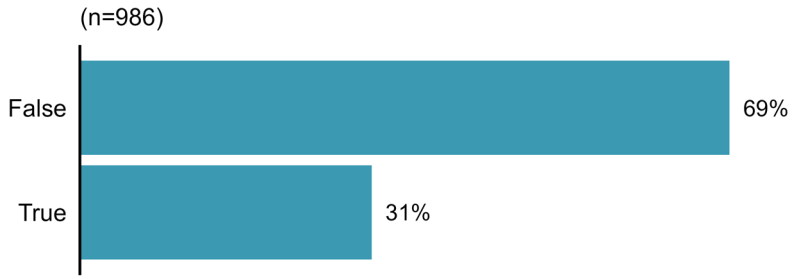
(IF Question Q4 = No, never)
Question Q6_2: Why did you not monitor the electricity prices? Multiple answers possible: Difficult finding infomation about electricity prices



(IF Question Q4 = No, never)
Question Q6_3: Why did you not monitor the electricity prices? Multiple answers possible: Difficult to understand what could be done



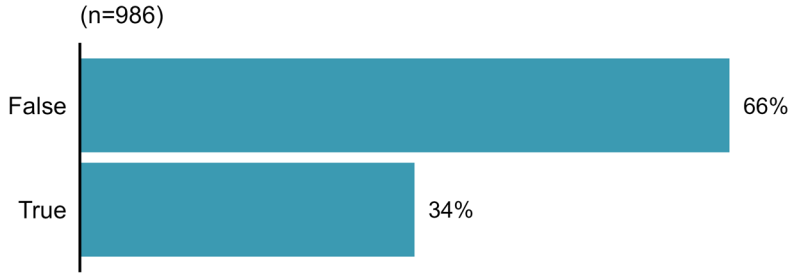
(IF Question Q4 = No, never)
Question Q6_4: Why did you not monitor the electricity prices? Multiple answers possible: Few possibilites to change consumption



(IF Question Q4 = No, never)
Question Q6_5: Why did you not monitor the electricity prices? Multiple answers possible: Potential savings too low



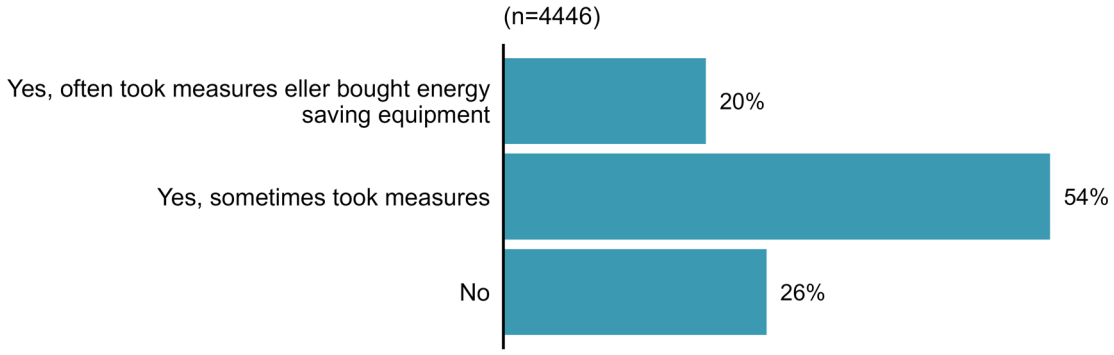
(IF Question Q4 = No, never)
Question Q6_6: Why did you not monitor the electricity prices? Multiple answers possible: Power bill too low



(IF Question Q4 = No, never)
Question Q6_7: Why did you not monitor the electricity prices? Multiple answers possible: Other:



Question Q7: Did you take any measures to decrease or move power consumption from hours with high prices this winter?



(IF Question Q7 = Yes, …)
Question Q8_1: Which measure did you implement? Multiple answers possible: Turned off electric heating in rooms when not in use



(IF Question Q7 = Yes, …)
Question Q8_2: Which measure did you implement? Multiple answers possible: Lowered the indoor temperature



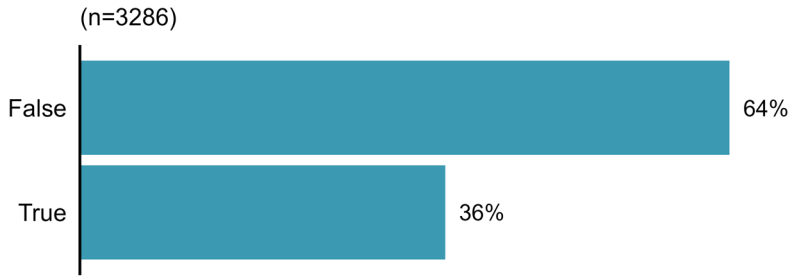
(IF Question Q7 = Yes, …)
Question Q8_3: Which measure did you implement? Multiple answers possible: Altered the schedule of timebased thermostats for electric heating



(IF Question Q7 = Yes, …)
Question Q8_4: Which measure did you implement? Multiple answers possible: Bought smart devices for controlling electric heating



(IF Question Q7 = Yes, …)
Question Q8_5: Which measure did you implement? Multiple answers possible: Used stove instead of electric heating



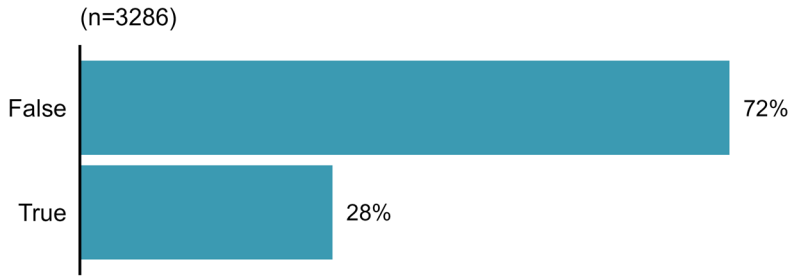
(IF Question Q7 = Yes, …)
Question Q8_6: Which measure did you implement? Multiple answers possible: Bought a heat pump



(IF Question Q7 = Yes, …)
Question Q8_7: Which measure did you implement? Multiple answers possible: Shifted dishwasher, washing machine or dryer use to different hours



(IF Question Q7 = Yes, …)
Question Q8_8: Which measure did you implement? Multiple answers possible: Shifted showering or bathing to different hours



(IF Question Q7 = Yes, …)
Question Q8_9: Which measure did you implement? Multiple answers possible: Shorter showers / fewers baths



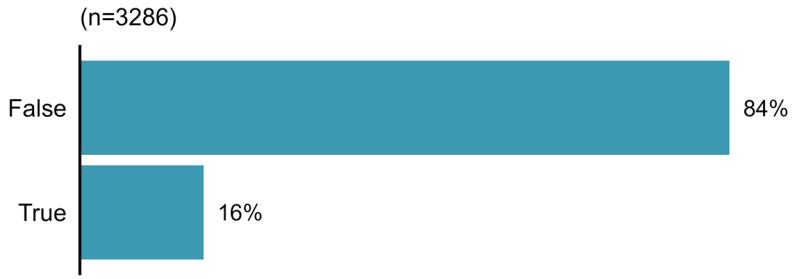
(IF Question Q7 = Yes, …)
Question Q8_10: Which measure did you implement? Multiple answers possible: Turned of the water heater



(IF Question Q7 = Yes, …)
Question Q8_11: Which measure did you implement? Multiple answers possible: Turned off lights or other small electronics



(IF Question Q7 = Yes, …)
Question Q8_12: Which measure did you implement? Multiple answers possible: Shifted electric car charging to different hours



(IF Question Q7 = Yes, …)
Question Q8_13: Which measure did you implement? Multiple answers possible: Installed smart car charger that charges when the prices are low



(IF Question Q7 = Yes, …)
Question Q8_14: Which measure did you implement? Multiple answers possible: Installed solar panels



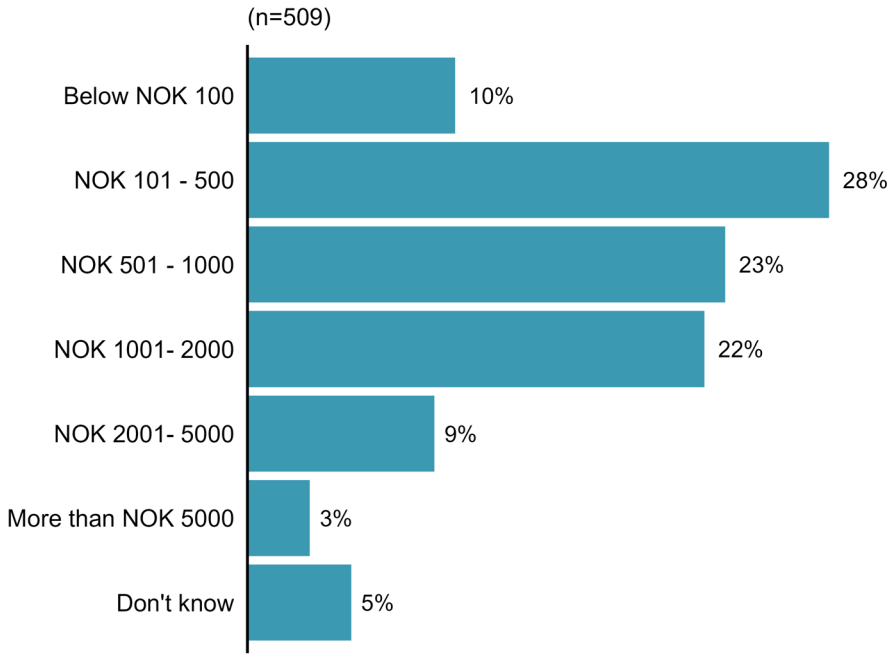
(IF Question Q7 = Yes, …)
Question Q8_15: Which measure did you implement? Multiple answers possible: Other:



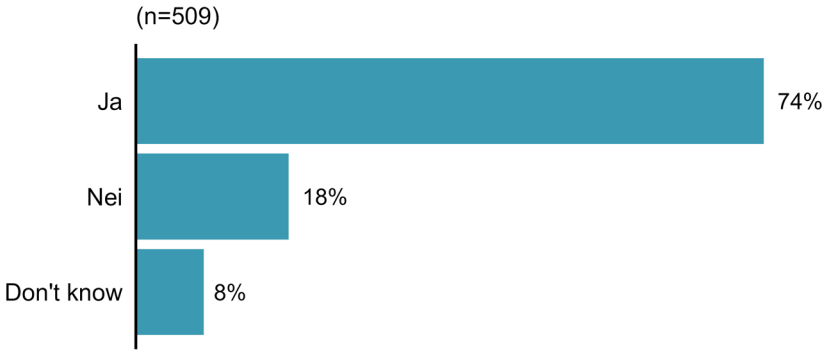
(IF Question Q7 = Yes, …)
Question Q9: Do you know how about much the household have saved on the power bill as a result of the measures?



(IF Question Q9 = Ja (Yes))
Question Q10: About how much has the household saved per month this winter as a result of the measures?



(IF Question Q9 = Ja (Yes))
Question Q11: Do you feel that the measures you implemented were worth the savings on the power bill?



(IF Question Q7 = Yes, …)
Question Q12_1: Why did you not take any measures? Multiple answers possible: No time or interest in taking measures



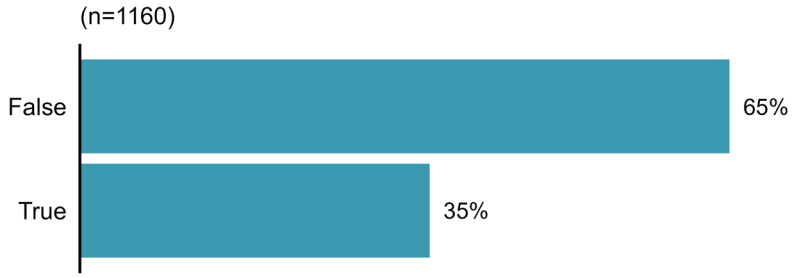
(IF Question Q7 = Yes, …)
Question Q12_2: Why did you not take any measures? Multiple answers possible: Difficult finding information about possible measures



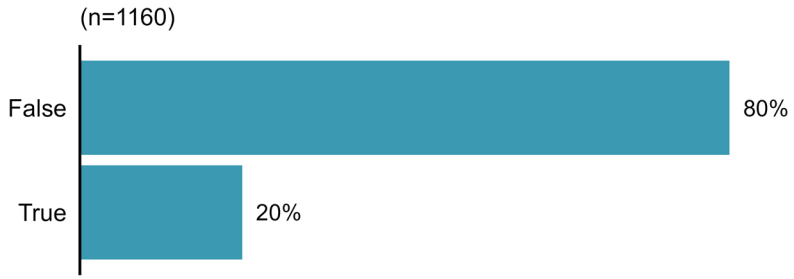
(IF Question Q7 = Yes, …)
Question Q12_3: Why did you not take any measures? Multiple answers possible: Difficult to understand which measures may be taken



(IF Question Q7 = Yes, …)
Question Q12_4: Why did you not take any measures? Multiple answers possible: Few possibilites to change consumption



(IF Question Q7 = Yes, …)
Question Q12_5: Why did you not take any measures? Multiple answers possible: Potential savings too low



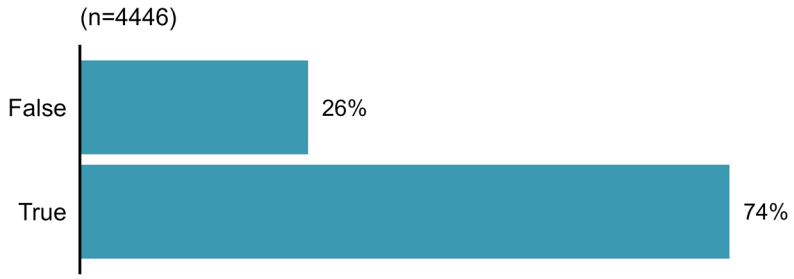
(IF Question Q7 = Yes, …)
Question Q12_6: Why did you not take any measures? Multiple answers possible: Power bill too low



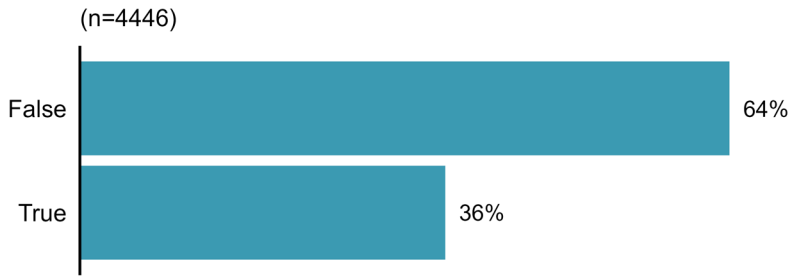
(IF Question Q7 = Yes, …)
Question Q12_7: Why did you not take any measures? Multiple answers possible: Other:



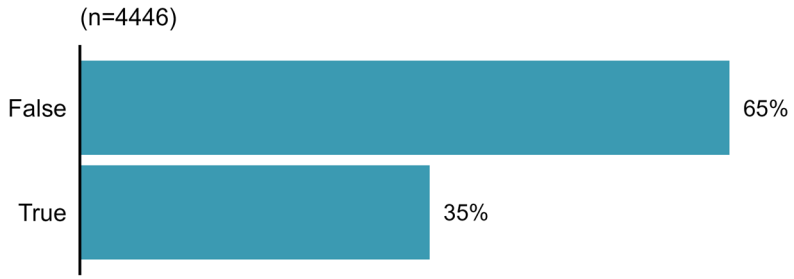
Question Q13_1: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Lower power bill



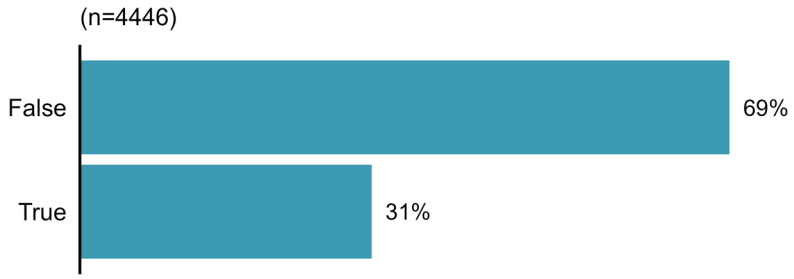
Question Q13_2: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Measures that are easy to do manually



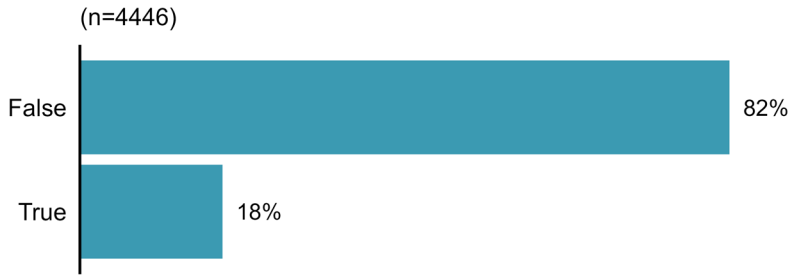
Question Q13_3: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: measures that happen automaticlly without having to think about it



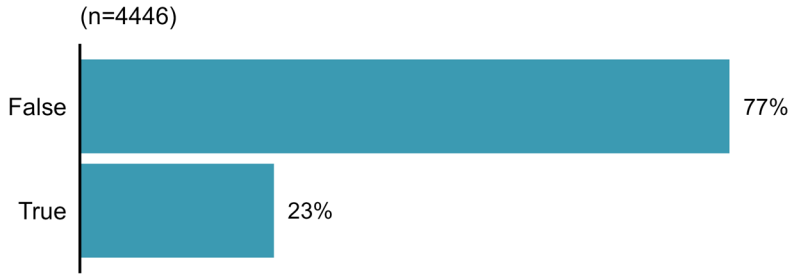
Question Q13_4: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Measure that do not affect comfort



Question Q13_5: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Testing new technology for controlling consumption



Question Q13_6: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Reducing the impact on nature and emissions from power production and grid



Question Q13_7: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Reducing the need for grid investments, thus reducing grid tariffs



Question Q13_8: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: If its a team effort



Question Q13_9: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: Other:



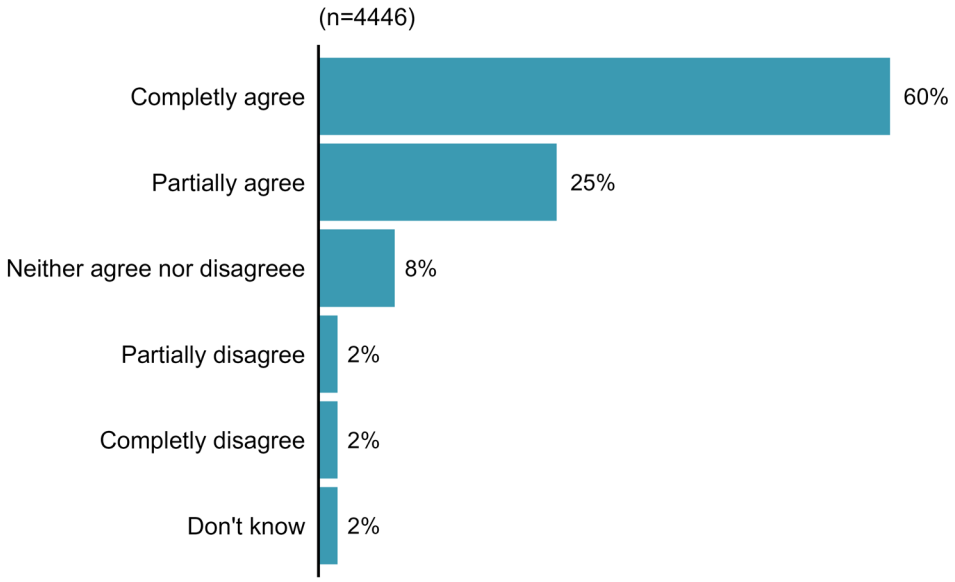
Question Q13_10: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: None of the above, I don't want to or cannot reduce my power consumption



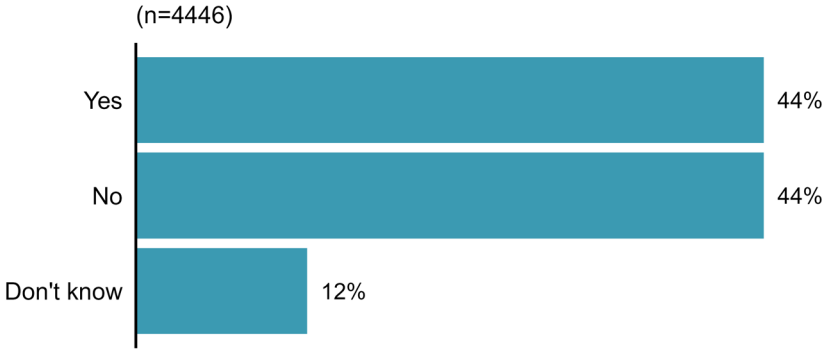
Question Q13_11: What motivates you to reduce you power consumption in high price hours? Multiple answers possible: I don't know



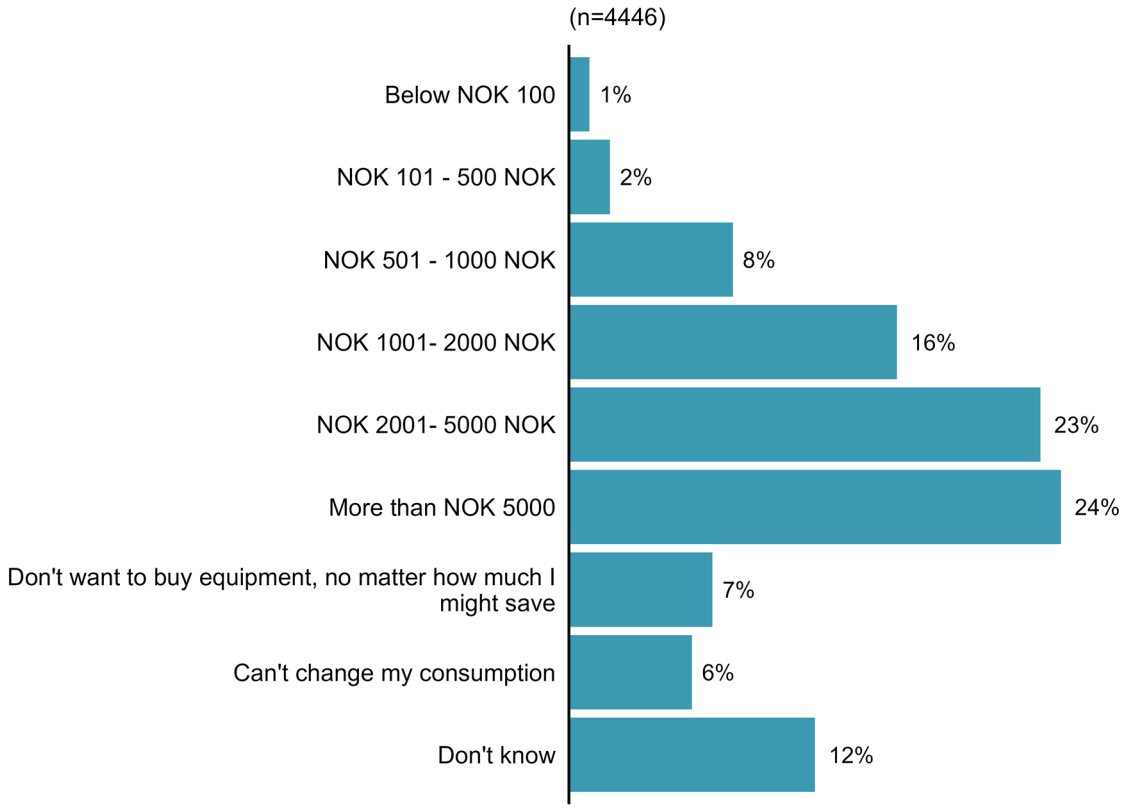
Question Q14: How much do you agree or disagree in the following statement? People who adjust their power consumption based on price should be able to save on their power bill



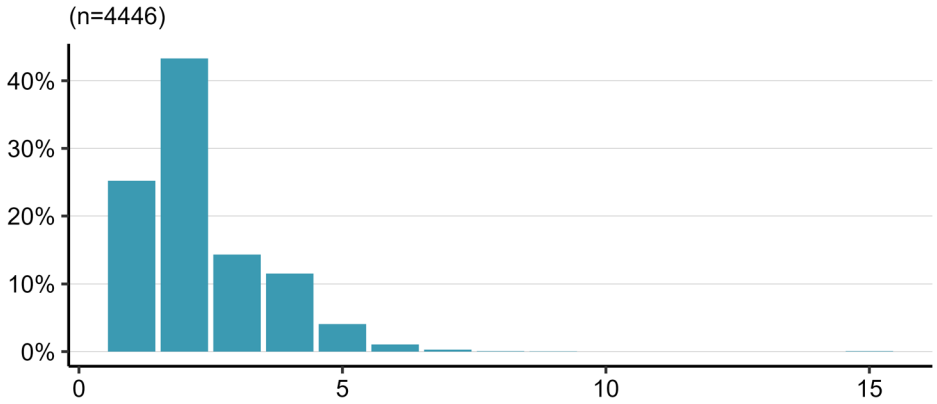
Question Q15: Would you or have you used a free information service that alerts you of high price hours the following day?



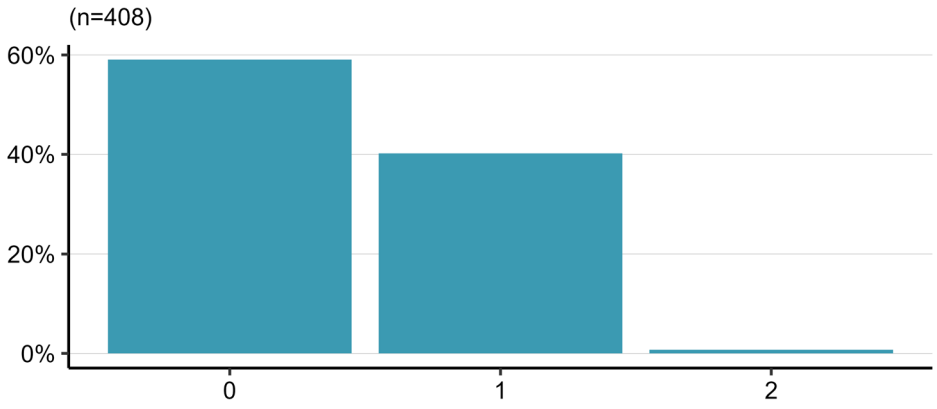
Question Q16: Imagine you can buy smart devices for 5000 NOK that will reduce you power bill by automatically shifting parts of you consumption away from high price hours - without reducing comfort. How much would you have to save every year to do it?



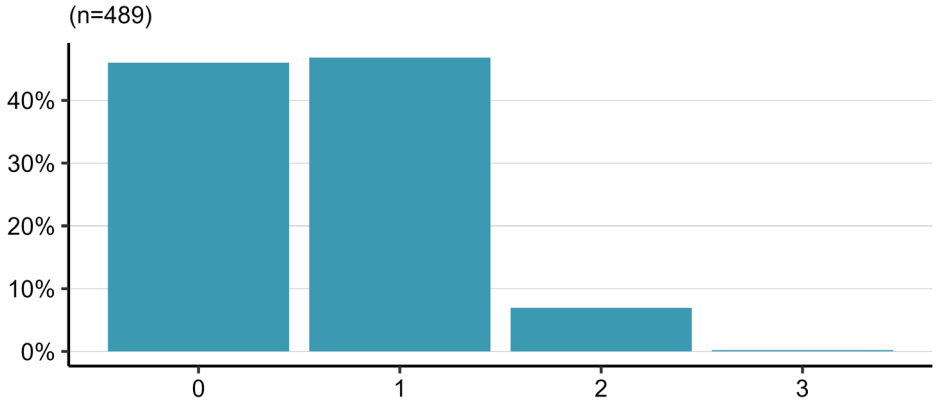
Question Q17: How many persons does you household consist of, including youself?



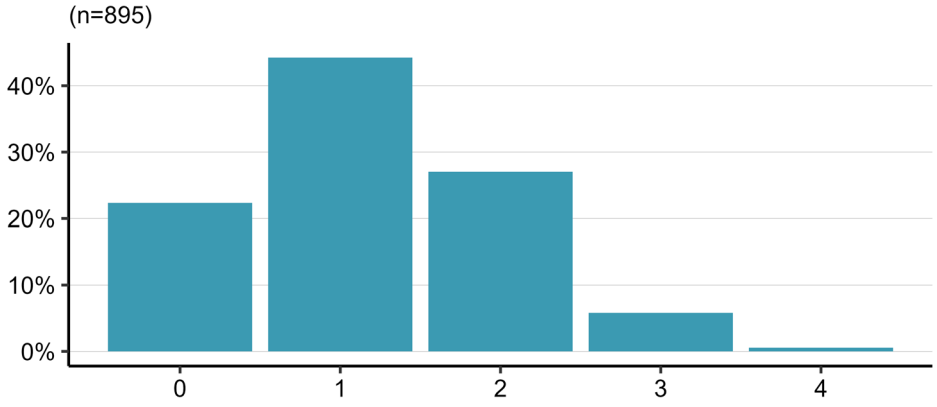
Question Q18_1: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 0-1 years



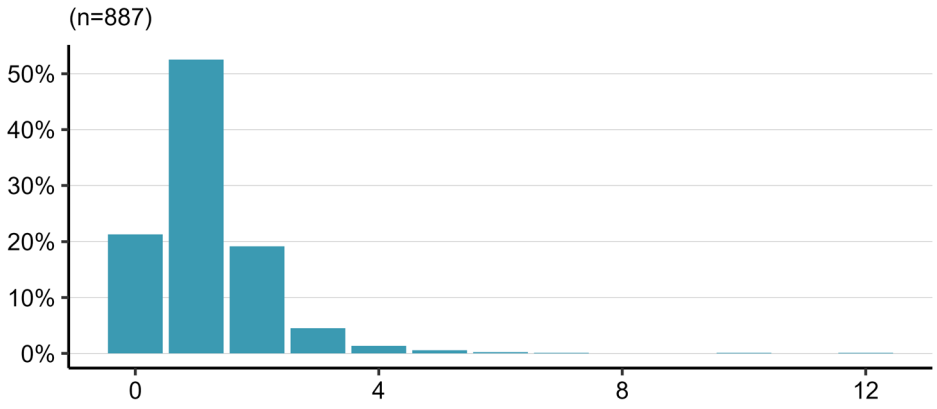
Question Q18_2: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 2-5 years



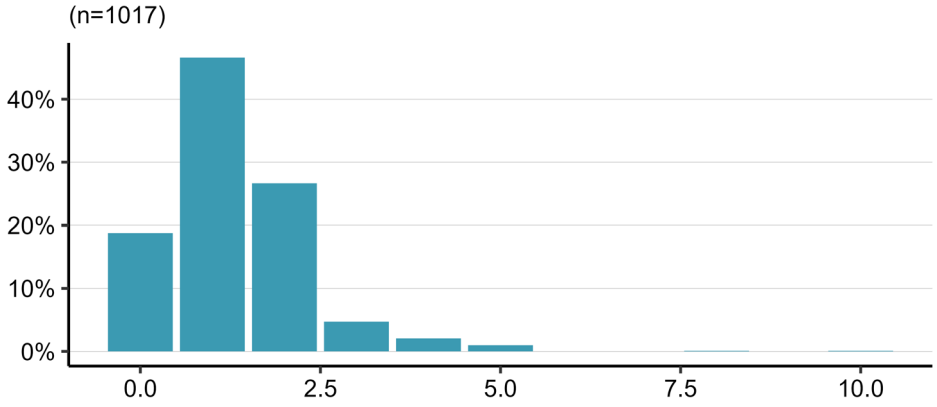
Question Q18_3: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 6-15 years



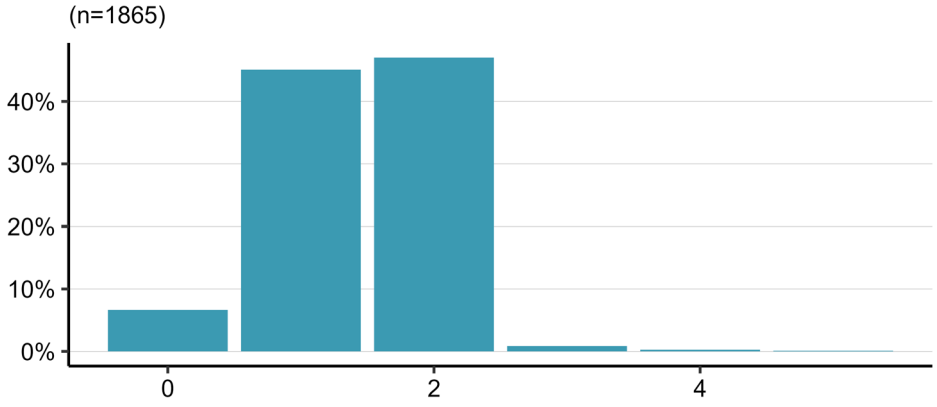
Question Q18_4: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 16-22 years



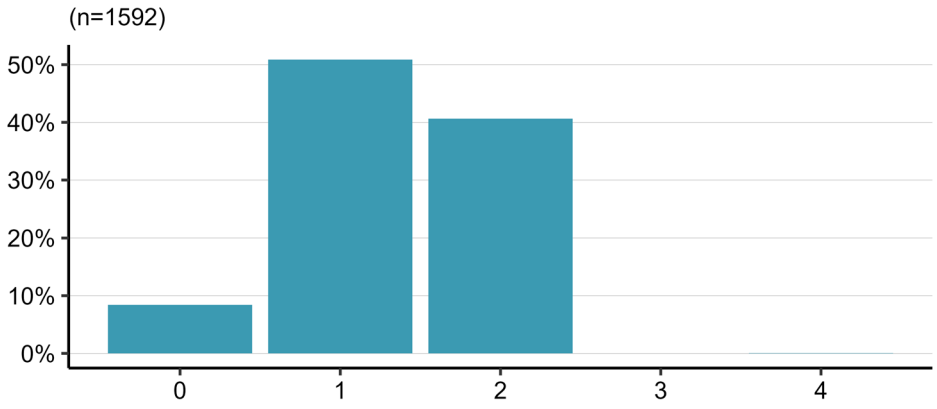
Question Q18_5: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 23-29 years



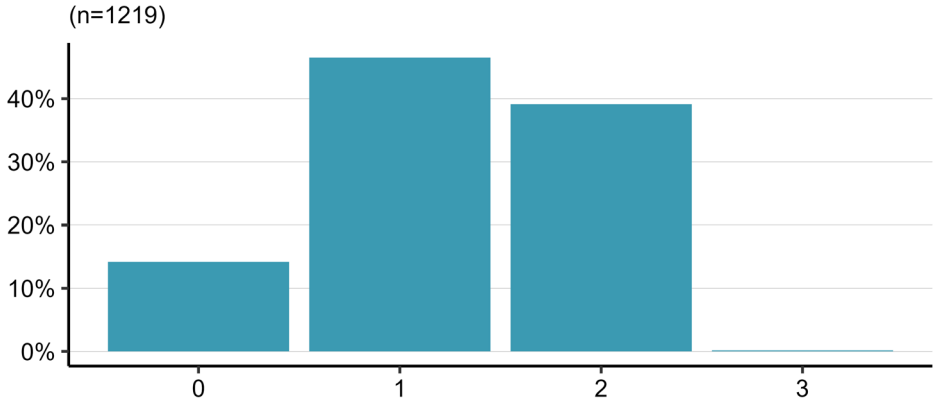
Question Q18_6: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 30-49 years



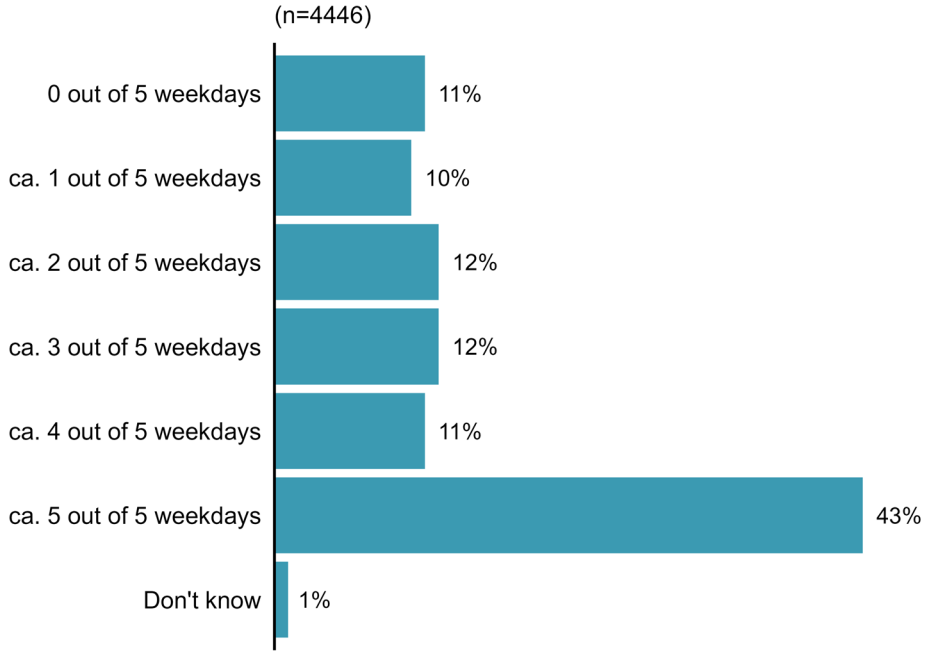
Question Q18_7: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 50-66 years



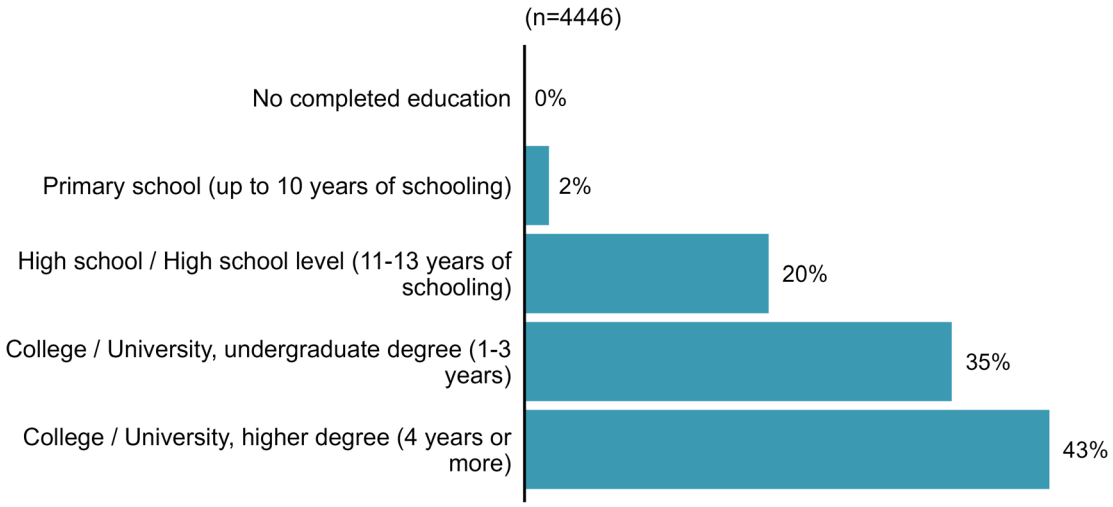
Question Q18_8: What age is the inhabitants of you household, including yourself? How many inhabitants are between: 67 years or older



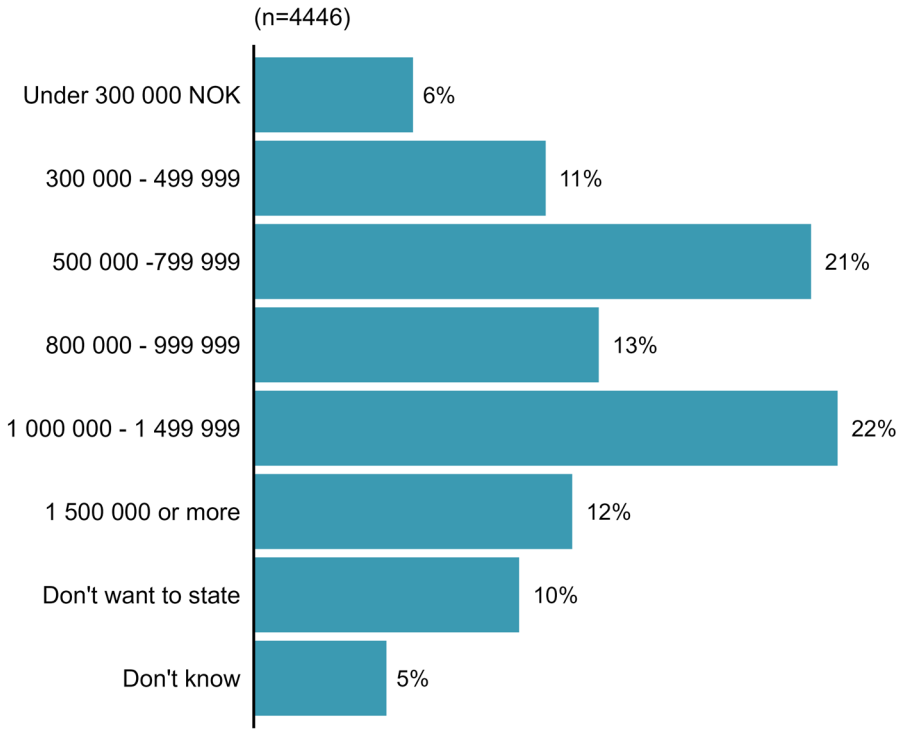
Question Q19: How many weekdays (Mon-Fri) in an average was there anyone home in the day (9-16) this winter (Nov-Mar)?



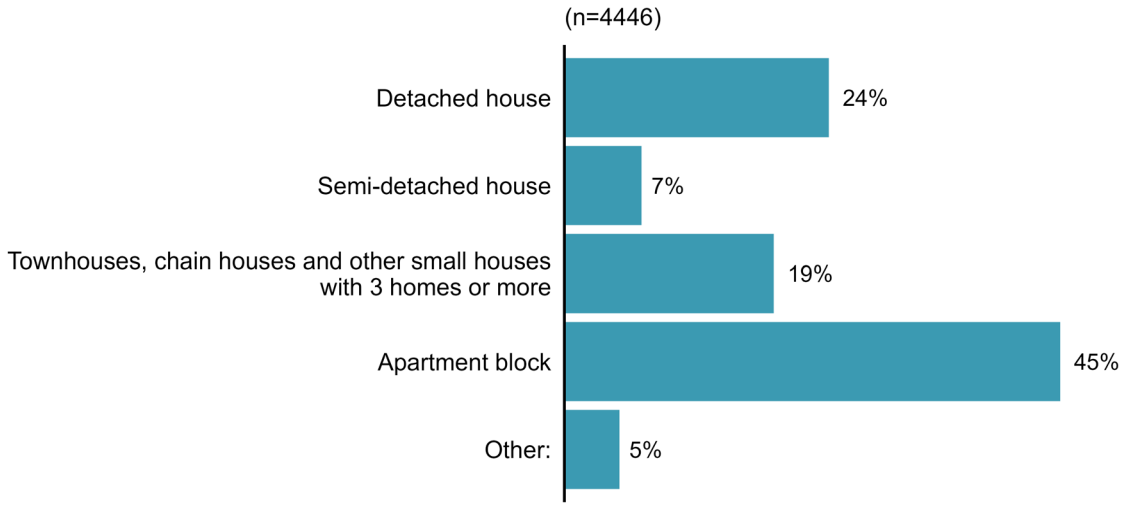
Question Q20: What is the highest education in the household?



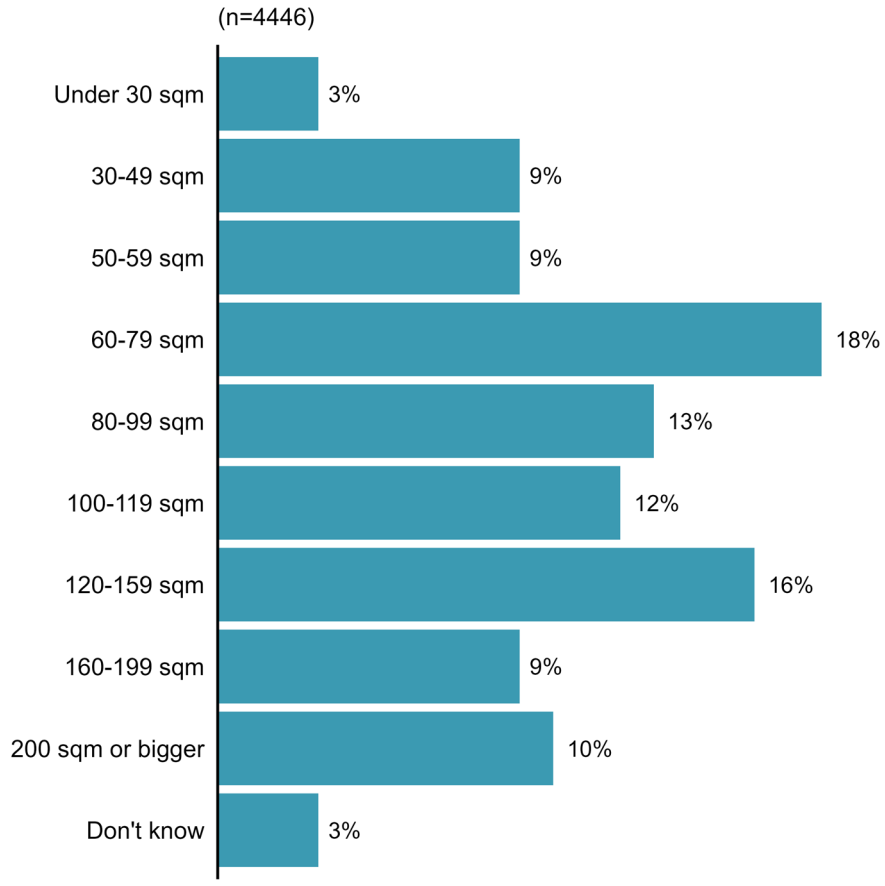
Question Q21: What is the combined gross income of the household?



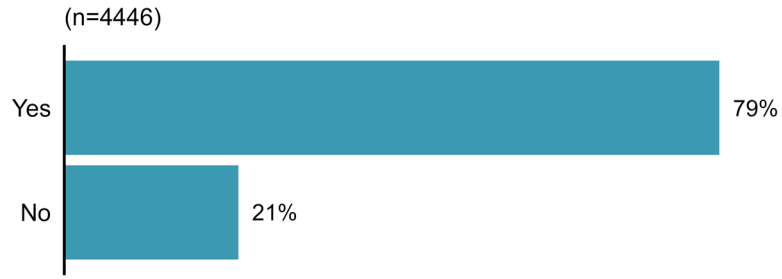
Question Q22: What type of residence do you live in?



Question Q23: How big is the residence?



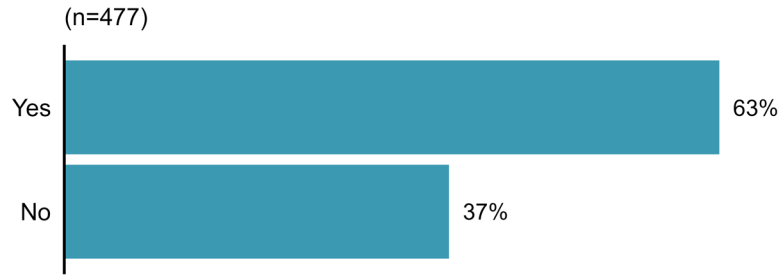
Question Q24: Do you own the residence?



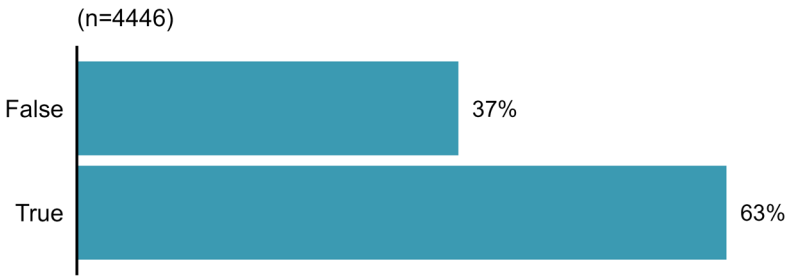
(IF Question Q24 = Yes)
Question Q25: Do you have a rental unit in the residence?



(IF Question Q25 = Yes)
Question Q26: Does the rental unit have its own power meter?



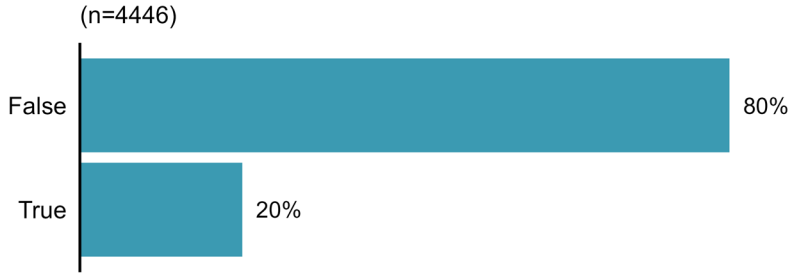
Question Q27_1: How is the residence heated? Multiple answers possible: Electric convection heater (wall mounted, free-standing, oil filled radiatior, etc.)



Question Q27_2: How is the residence heated? Multiple answers possible: Electric floor heating (heater cable, heating foil, etc.)



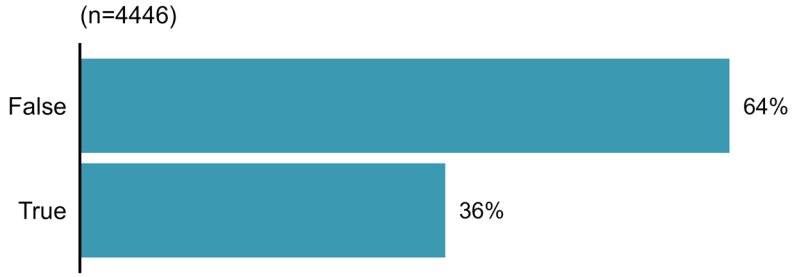
Question Q27_3: How is the residence heated? Multiple answers possible: Air-air or air-water heat pump



Question Q27_4: How is the residence heated? Multiple answers possible: Ground heating



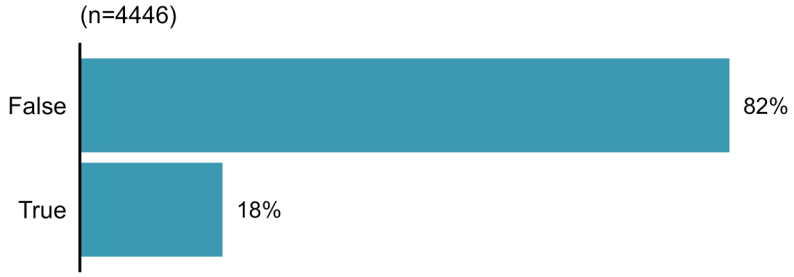
Question Q27_5: How is the residence heated? Multiple answers possible: Fireplace / wood stove



Question Q27_6: How is the residence heated? Multiple answers possible: Oil-/kerosine-/gas-/bio-fired heaters



Question Q27_7: How is the residence heated? Multiple answers possible: District heating or other collective building heating solution



Question Q27_8: How is the residence heated? Multiple answers possible: Waterborne heating that is heated electrically in the residence



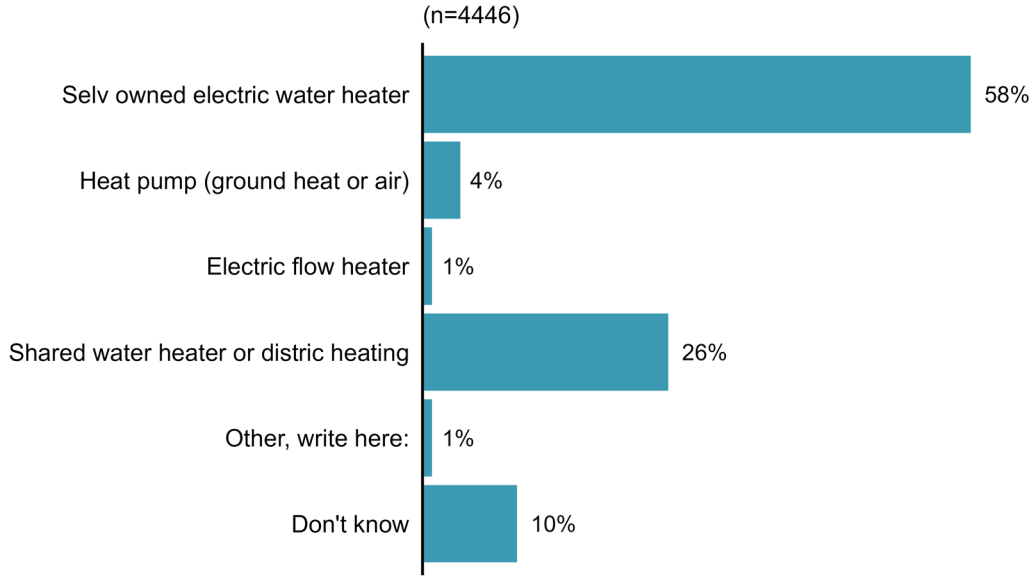
Question Q27_9: How is the residence heated? Multiple answers possible: Other:



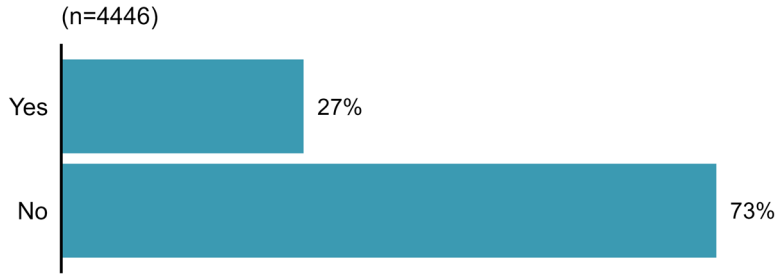
Question Q27_10: How is the residence heated? Multiple answers possible: Don't know



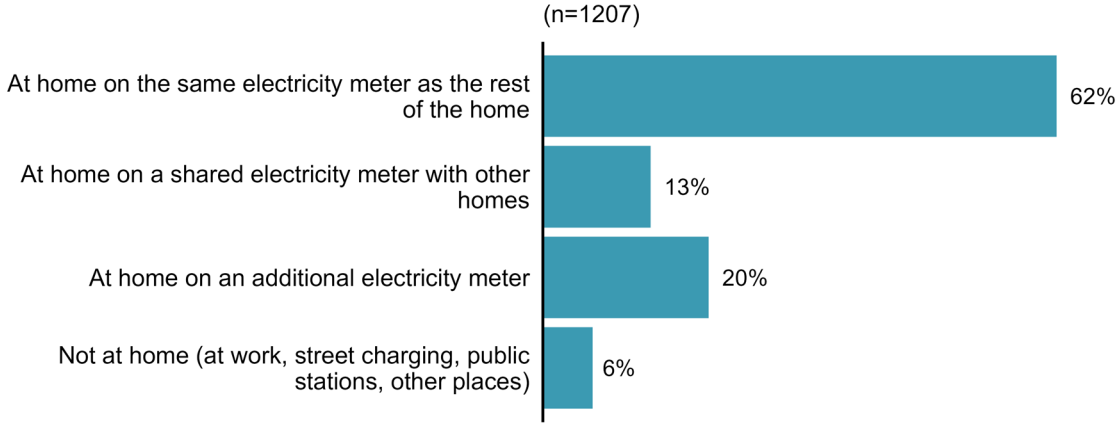
Question Q28: How is the tap water heated?



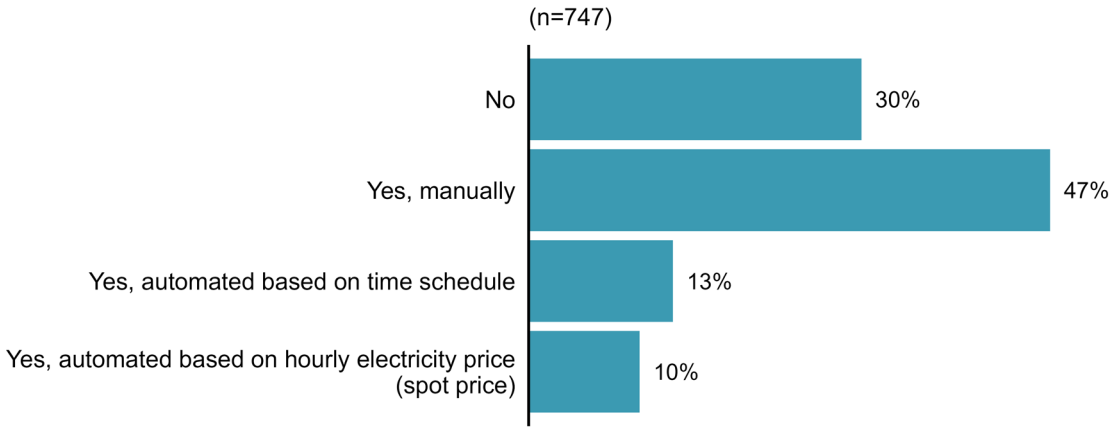
Question Q29: Do you own one or more electric car that is at least sometimes charged at home?



(IF Question Q29 = Yes)
Question Q30: How is the car or cars normally charged?



(IF Question Q30 = At home on the same electricity meter …)
Question Q31: Do you control the car charging to avoid hours with high prices?



Question Q32: What type of power contract do you have?

